

NXC5014 Elastomers

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1. Introduction of Rubber & Elastomer

■ Definition

- **Elastomer**, A macromolecular material which returns to its initial shape rapidly after a weak stress.
- **Rubber**, A **insoluble** but can **swell** elastomer, and can not easily reshaped by heat and pressure.

1.1. Polyisoprene

- **Hevea NR**, *cis*-1,4-polyisoprene
- **Gutta or Percha NR**, *trans*-1,4-polyisoprene

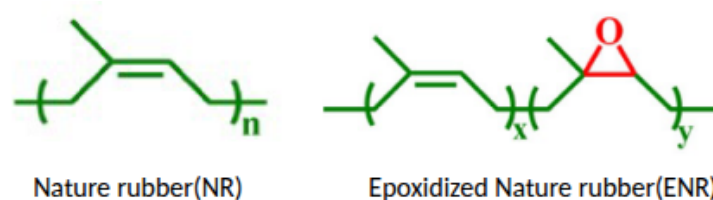
Hevea NR has advantages:

- $T_f = 130^\circ\text{C}$, $T_{d-set} = 200^\circ\text{C}$, and $T_{d-max} = 270^\circ\text{C}$.
- NR contains small amounts of fatty acids and proteinaceous residues that promote sulphur vulcanisation and possibly strain induced crystallisation. **Self-reinforcing effect**.
- Has Excellent properties, including processing, mechanical, gas permeability and electron insulation.

However, Hevea NR is **Unsaturated & Non-polar**, easy to be degraded.

■ Epoxidized Natural Rubber, ENR

Epoxidized natural rubber (ENR), a chemically modified derivative of NR, can be easily and efficiently produced through *in-situ* epoxidation of the NR latex.



1.2. Naming Conventions

Usually designate polymers by **P** followed by first letters indicating the monomer

Rubbers (raw or cured) are usually designated by letters indicating the monomer(s) followed by **R**

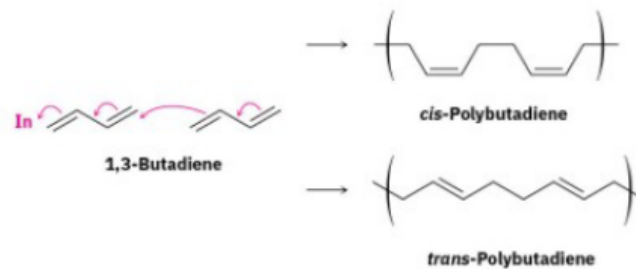
■ Block Copolymer, BCP

1. Homopolymer
2. Alternating Copolymer
3. Random Copolymer
4. Block Copolymer
5. Graft Copolymer

1.3. Rubber Types

■ Polybutadiene Rubber, BR

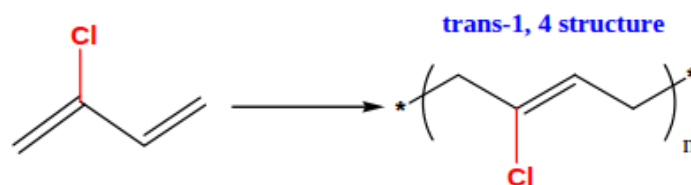
A conjugated diene monomer



Ziegler-Natta BR, High *cis*-isomer concentration, it mimics 100% *cis*-1,4 structure of NR.

■ Polychloroprene, CR

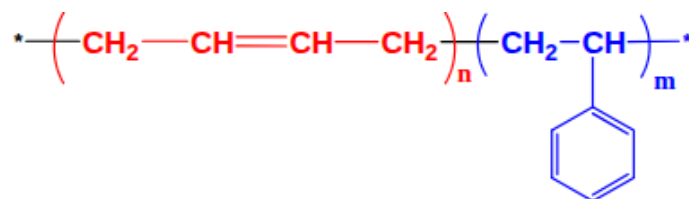
- C–Cl dipoles enhance interchain interaction and crystallization
- CR has good adhesion to polar substrates
- The electron-withdrawing Cl atom protect the double bond from O₂ and O₃.



Applications include wire, cable, hose, and some mechanical goods.

■ Styrene-Butadiene Rubber

Also name SBR, a copolymer of styrene and butadiene.



SBR is the most widely used synthetic elastomer.

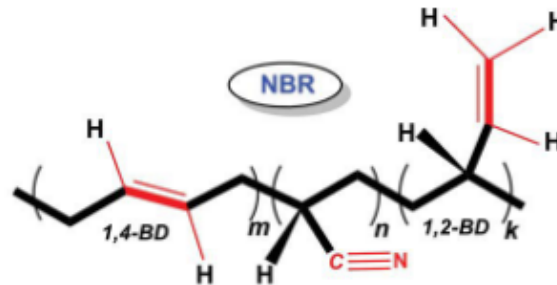
SBR is synthesised via free-radical as an emulsion(High *T* cause highly branched, so often in *cold* or 5°C) or anionically in solution.

It does not strain crystallise. Therefore reinforcing fillers such as *silica* or *carbon black* have to be used to impart adequate properties

■ Acrylonitrile-Butadiene Rubber, NBR

Polarity in NBR is introduced by the polar monomer, acrylonitrile, excellent fuel and oil resistance.

With increased acrylonitrile content, there is an increase in T_g , reduction in resilience, lower die swell, decreased gas permeability, increased heat resistance, and increased strength.



Higher ACN Ratio

Lower crosslink density, **Larger** compression set, **Higher** glass transition temperature T_g .

■ Ethylene-Propylene Rubber, EPR or EPDM

EPRs and EPDMs have excellent resistance to weathering and good heat stability. Excellent electronic insulation and Elasticity (only less than NR and BR).

But:

1. Slow cross-linking
2. Low tack and adhesion
3. Low crystalline and self-reinforcement

Applications include roofing, seals, gaskets, and hose, widely in aerospace.

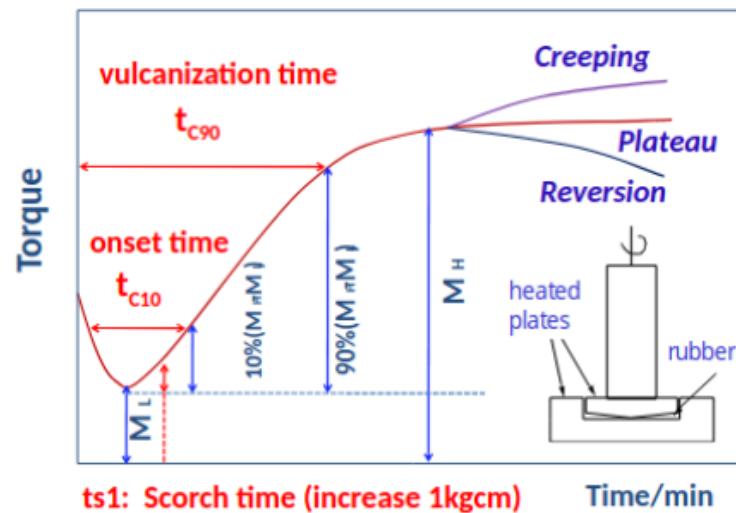
1.4. Entropic Elasticity of Rubber

Unlike metals and ceramics whose elasticity derives from **internal energy** changes (bond stretching/bending), rubber elasticity is primarily **entropic** in nature.

- **Thermodynamic basis:** $\Delta G = \Delta H - T\Delta S$. For an ideal elastomer, $\Delta H \approx 0$, so the retractive force arises entirely from the entropy term $-T\Delta S$.
- **Molecular origin:** Polymer chains in the unstretched state adopt random-coil conformations with **maximum conformational entropy**. Stretching reduces the number of accessible conformations, lowering entropy ($\Delta S < 0$).
- **Retractive force:** $f = \left(\frac{\partial G}{\partial L}\right)_T = -T \left(\frac{\partial S}{\partial L}\right)_T$. The restoring force is **proportional to absolute temperature** T — opposite to metals, where stiffness decreases on heating.
- **Gough–Joule effect:** A stretched rubber band **contracts** when heated (and expands when cooled), directly demonstrating the entropic origin of elasticity.
- **Requirements for entropic elasticity:**
 1. Long, flexible polymer chains (high molecular weight)
 2. Light cross-linking to prevent viscous flow (network structure)
 3. $T > T_g$ (chains must be mobile above the glass transition)

2. Chemistry & Physics of Synthetic Rubbers

2.1. Vulcanizing/Curing Mechanism



■ Accelerators

Accelerator increase the speed of vulcanization and to permit vulcanization to proceed at **lower temperature** and with **greater efficiency**.

Accelerator also Decreases the Quantity of Sulphur necessary for vulcanization and thus improving 'aged' properties of the rubber vulcanizates.

The use of secondary accelerators increases the speed of vulcanization substantially but at the expense of scorch safety. The dosages of the secondary accelerators are generally between 10–40% of the primary accelerator.

■ Activators

Metal oxide(main) and fatty acid.

Activate cure and improve the efficiency of sulfur based cure systems.

2.2. Rubber Mixing

To improve the performance and cost saving, **Various Agents** should be added into rubber.

Rubber mixing is mix the agents and rubber by mechanical force.

2.3. Plastication of Rubber

Increasing plasticity of rubber by mechanical or chemical method to meet the various processing requirements of rubber.

Including cold plastication, heat plasticization, and **peptizer (or plasticizer)**.

■ Mechanical Plastication

The probability of breaking molecular chain is $\propto$ mechanical worked, and inversely $\propto$ the temperature of rubber.

However, the free radicals can re-combine the chain rapidly, so only mechanical force is not enough.

■ Oxygen

O₂ can degrade free radicals, and react with rubber to make them degradation.

Cold and Heat Plasticization

In cold plasticization, the breaking of chain is mainly caused by mechanical force.

In heat plasticization, the main reason is oxygen degradation.

2.4. Reinforcement

Particulate fillers can increase the strength of an amorphous rubber. The filler should have high specific surface area.

■ Carbon Black

Primary structure of carbon black, about 300 nm.

Secondary structure, the space network formed by van der Waals force. It should be crushed by mechanical force.

Oil adsorption

Crude but effective method to determine how much oil is required to fill up structural voids in carbon black aggregates. Units are $^3g^{-1}$.

The higher oil adsorption, the higher structure of CB.

2.5. Anti-Degradants

Aromatic Amines, Chemical antiozonants are required to protect all of the external vulcanized rubber components from ozone-induced cracking.

3. Characterization

3.1. Thermogravimetric Analysis (TGA)

TGA measures amount and rate of weight change vs. temperature or time in a controlled atmosphere.

Characterizes materials that exhibit weight loss or gain due to decomposition, oxidation, or dehydration

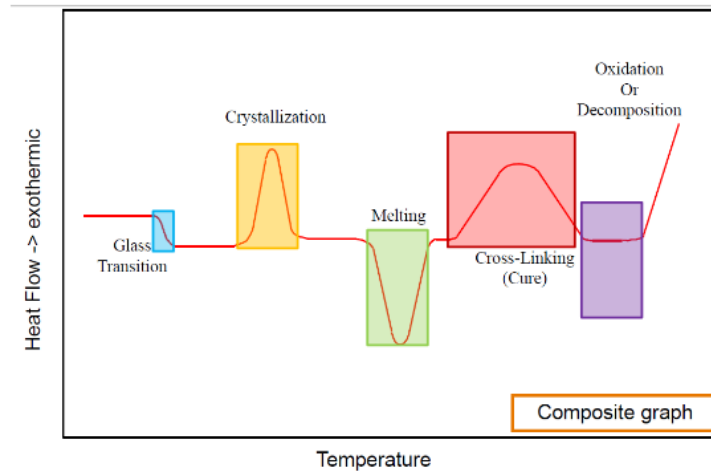
Cautions

TGA are all kinetic processes, which means it dependent on absolute temperature and time spent

at that temperature.

3.2. Differential Scanning Calorimetry(DSC)

Differential scanning calorimetry (DSC) measures the energy absorbed(endothermic) or produced (exothermic) as a function of time or temperature.



It is used to characterize melting, crystallization, resin curing, loss of solvents, and other processes involving an energy change.

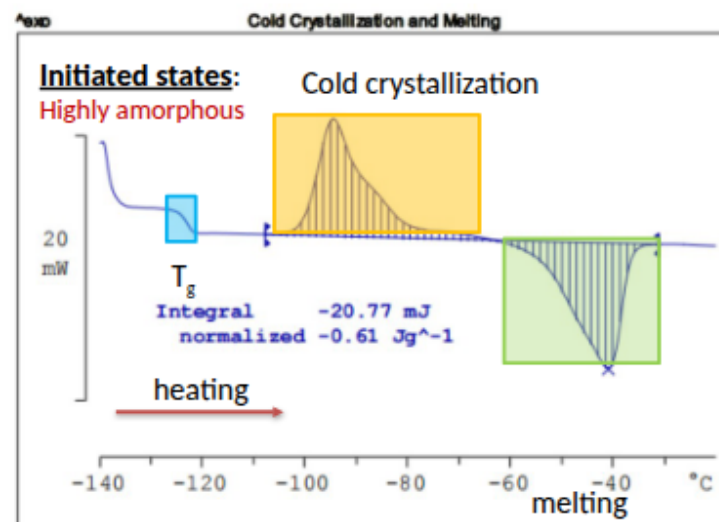
DSC may also be applied to processes involving a change in heat capacity, such as the glass transition

Cold Crystallization

When a polymer is rapidly cooled (quenched) from the melt to below T_g , chains are frozen in an amorphous state.

Upon reheating above T_g , chains gain mobility and reorganize into crystalline regions — this exothermic peak is **cold crystallization** (T_{cc}).

It occurs between T_g and T_m , indicating the polymer was not fully crystallized during cooling.



3.3. Dynamic mechanical analysis(DMA)

Measures the mechanical properties of a sample as it is deformed over a range of stress, strain, time and temperature.

4. Structure & Property of Rubber

4.1. Statistical Elasticity Theory

Consider a single polymer chain at equilibrium:

- Random coil
- n segments of length l
- Total contour length is nl
- The distance between chain ends is r

The chain position:

- **Freely Jointed Chain**, $(\bar{r}^2)^{\frac{1}{2}} = n^{\frac{1}{2}} l$
- **Freely Rotating Chain**, $(\bar{r}^2)^{\frac{1}{2}} = (2n)^{\frac{1}{2}} l$

From statistical thermodynamics, entropy

$$S = k \ln \Omega$$

Then,

$$S = c - kb^2 r^2 \quad b^2 = \frac{3}{2nl^2}$$

4.2. Neo-Hookean Stored Energy Function

- **Stored energy function** (incompressible):

$$W = \frac{\mu}{2}(I_1 - 3), \quad I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$$

where λ_i are principal stretch ratios, I_1 is the first invariant of the Cauchy–Green deformation tensor.

- **Shear modulus**: $\mu = nkT = \frac{\rho RT}{M_c}$, where n is chain density, k Boltzmann constant, T absolute temperature, ρ density, R gas constant, M_c molecular weight between crosslinks.
- **Incompressibility constraint**: $\lambda_1 \lambda_2 \lambda_3 = 1$, rubber deforms at constant volume.

4.3. Hyperelastic Models

Hyperelastic means:

- No loss of energy
- No time dependence of the behaviour
- No path dependence on final stress and strains.

Hyperelastic models include:

- **Neo-Hookean:** $W = \frac{G}{2}(I_1 - 3)$. Simple one-parameter model; good for small/medium strains but poor at large strains.
- **Mooney–Rivlin:** $W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$, where $I_2 = \lambda_1^{-2}\lambda_2^{-2} + \lambda_2^{-2}\lambda_3^{-2} + \lambda_3^{-2}\lambda_1^{-2}$. Better than Neo-Hookean for larger strains.
- **Rivlin series:** $W = \sum_{i,j=0}^{\infty} C_{ij}(I_1 - 3)^i(I_2 - 3)^j$.
- **Arruda–Boyce:** accounts for finite chain extensibility; parameters G and λ_m .
- **Gent:** $W = -\frac{G}{2}(I_m - 3) \ln \left(1 - \frac{I_1 - 3}{I_m - 3}\right)$; captures strain hardening.
- **Yeoh:** $W = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3$; only I_1 needed.

Model Selection

If possible use a strain-energy function of I_1 only. Then a single characterization test is sufficient.

■ Uniaxial Tension

For incompressible uniaxial tension ($\lambda_1 = \lambda$, $\lambda_2 = \lambda_3 = \lambda^{-1/2}$):

- Neo-Hookean engineering stress: $\sigma = G(\lambda - \lambda^{-2})$.
- True stress: $t = \sigma\lambda = G(\lambda^2 - \lambda^{-1})$.

■ Limitations of Neo-Hookean Model

- **Medium strains:** network imperfections (loose ends, closed loops, trapped entanglements) cause slight overestimation.
- **High strains:** Gaussian assumption fails; finite extensibility and strain-induced crystallisation become important.

4.4. Inelastic Behaviour

Real rubber is not always hyperelastic. Inelasticity arises from:

1. Viscoelastic effects (temperature and rate dependence, proximity to T_g).
2. Presence of reinforcing fillers.

Consequences:

- Time/rate dependence
- Path dependence
- Energy loss (hysteresis)

■ Glass Transition

T_g is not a true thermodynamic transition; it is a change in the timescale needed for chain segments to reach equilibrium. Fast deformation at high T is equivalent to slow deformation at low T .

■ Dynamic Mechanical Analysis

A sinusoidal deformation gives:

- Storage modulus E' : elastic component.
- Loss modulus E'' : viscous component.
- Loss factor $\tan \delta = E''/E'$: peaks around T_g .

■ Time–Temperature Superposition (TTS)

WLF equation shifts isotherms along the frequency axis to produce a master curve:

$$\log a_T = \frac{-C_1(T - T_{ref})}{C_2 + (T - T_{ref})}$$

Typical universal constants: $C_1 = 17.4$, $C_2 = 51.6$ (when $T_{ref} = T_g$).

■ Static Inelasticity

- **Creep**: deformation under constant stress. Creep rate tracks $\tan \delta(T, \omega)$.
- **Stress relaxation**: stress decay under constant strain. Physical relaxation (network reorientation) dominates initially; chemical relaxation (bond scission) dominates at high T /long times.

■ Mullins Effect

Cyclic stress softening in filled rubber. The equilibrium curve depends on the maximum previously applied strain. Softening can recover over time in the unstrained state.

■ Payne Effect

Dynamic modulus decreases with increasing strain amplitude in filled rubber. Mechanisms include filler-network breakdown/reformation, polymer adsorption/desorption at filler surface, and disentanglement.

4.5. Failure: Aging and Swelling

Rubber properties change in service due to microstructural changes:

- Thermal aging
- Ozone/oxygen attack
- Swelling by liquids
- Gas permeation and chemical attack

■ Thermal Aging

High temperature causes:

1. Continued curing (if under-cured) $\rightarrow$ stiffer.
2. Network modification: crosslink scission/interchange/insertion, main-chain scission, extra-network material.

Bond energies: C–C (~ 352 kJ/mol) strongest; polysulfidic C–S_x–C (~ 268 kJ/mol) weakest. Interestingly, weaker crosslinks can yield stronger vulcanisates through yielding and reformation.

■ Ozone/Oxygen Ageing

- Diffusion-controlled; mainly affects surface layers.
- Ozone cracking occurs only under tensile strain.
- Protected by antioxidants and waxes that bloom to the surface.

■ Swelling

Crosslinked rubber swells when solvent mixing free energy overcomes the elastic resistance of the network. Equilibrium is described by the Flory–Rehner equation:

$$\ln(1 - v_r) + v_r + \chi v_r^2 + v_e V_1 v_r^{1/3} = 0$$

where v_r is the volume fraction of rubber, χ the polymer–solvent interaction parameter, v_e the effective crosslink density, and V_1 the solvent molar volume.

Solubility parameter δ : when polymer and solvent δ are close, swelling is strong.

Effect of Filler

Filler does not swell and strongly interacts with polymer, so true crosslink density cannot be calculated directly from swelling experiments of filled compounds.

■ Gas Saturation and Chemical Attack

- Gas solubility follows Henry's law: $c = sP$.
- H_2S can crosslink HNBR, increasing modulus and T_g while reducing elongation.
- Lifetime prediction by accelerated kinetics is complicated because T affects reaction, solubility and diffusion simultaneously.

4.6. Strength of Rubber

Rubber usually fails by crack growth from flaws, not by uniform tensile rupture. The crack-growth characteristics determine strength.

■ Fracture Energy

Energy balance for crack growth:

$$\delta W \geq \delta U_{el} + G_c t \delta a$$

G_c (or T) is the energy absorbed per unit crack area ($J m^{-2}$). It is a material property independent of specimen geometry.

■ Tearing Tests

- Pure shear: $T = Wh$ (W = strain-energy density, h = height).
- Trouser tear: $T = \frac{2\lambda F}{t} - w_0 W$.

■ Factors Affecting Strength

- Viscoelasticity: more viscoelastic rubber generally has higher G_c .
- Crystallisation: NR strain-crystallises at the crack tip, blunting cracks; less rate/temperature sensitive.
- Fillers: increase hysteresis and cause crack branching/blunting.

■ Void Inflation and Rapid Gas Decompression

- Hydrostatic tension causes cavitation.
- For a spherical void in Neo-Hookean material, critical inflation pressure $P_c = \frac{5G}{2}$.
- High initial modulus is therefore desirable for RGD-resistant seals.

4.7. Friction of Rubber

Rubber friction does not follow Coulomb's law ($\mu = F/N$ constant). It depends on:

- Surface roughness of rubber and substrate
- Adhesion
- Viscoelastic properties (temperature, structure, filler)
- Lubricants and humidity

■ Friction Components

$$F_{total} = F_{adhesion} + F_{deformation} + F_{viscous} + F_{cohesion}$$

- **Adhesion:** dominates on smooth dry surfaces; involves stick-slip and Schallamach waves.
- **Deformation (hysteresis):** dominates on rough surfaces; cyclic deformation of rubber by asperities.
- **Viscous:** due to fluid films, reduces friction.
- **Cohesion:** from tearing rubber.

■ Pressure and Velocity Dependence

- Friction coefficient decreases with increasing normal pressure because the real contact area grows sub-linearly.
- Velocity and temperature dependence can be described by WLF time–temperature superposition.

■ Wet Friction and Hydroplaning

- Water can reduce friction by 20–30%.
- A thick water film causes hydroplaning, lifting the rubber from the substrate.
- On ice, the surface is near its melting point, making friction behaviour very sensitive to pressure, speed and temperature.

■ Tyre Tread Design

- Good wet grip: high $\tan \delta$ at frequencies corresponding to road roughness.
- Low rolling resistance: low $\tan \delta$ at frequencies corresponding to tyre rotation.